Lecture 10 (Cones)

February 18, 2026

A cone is a special type of set. Cones have specific properties that make them useful when working with dual methods for optimization. We will define cones today and show some of their properties.

Cones

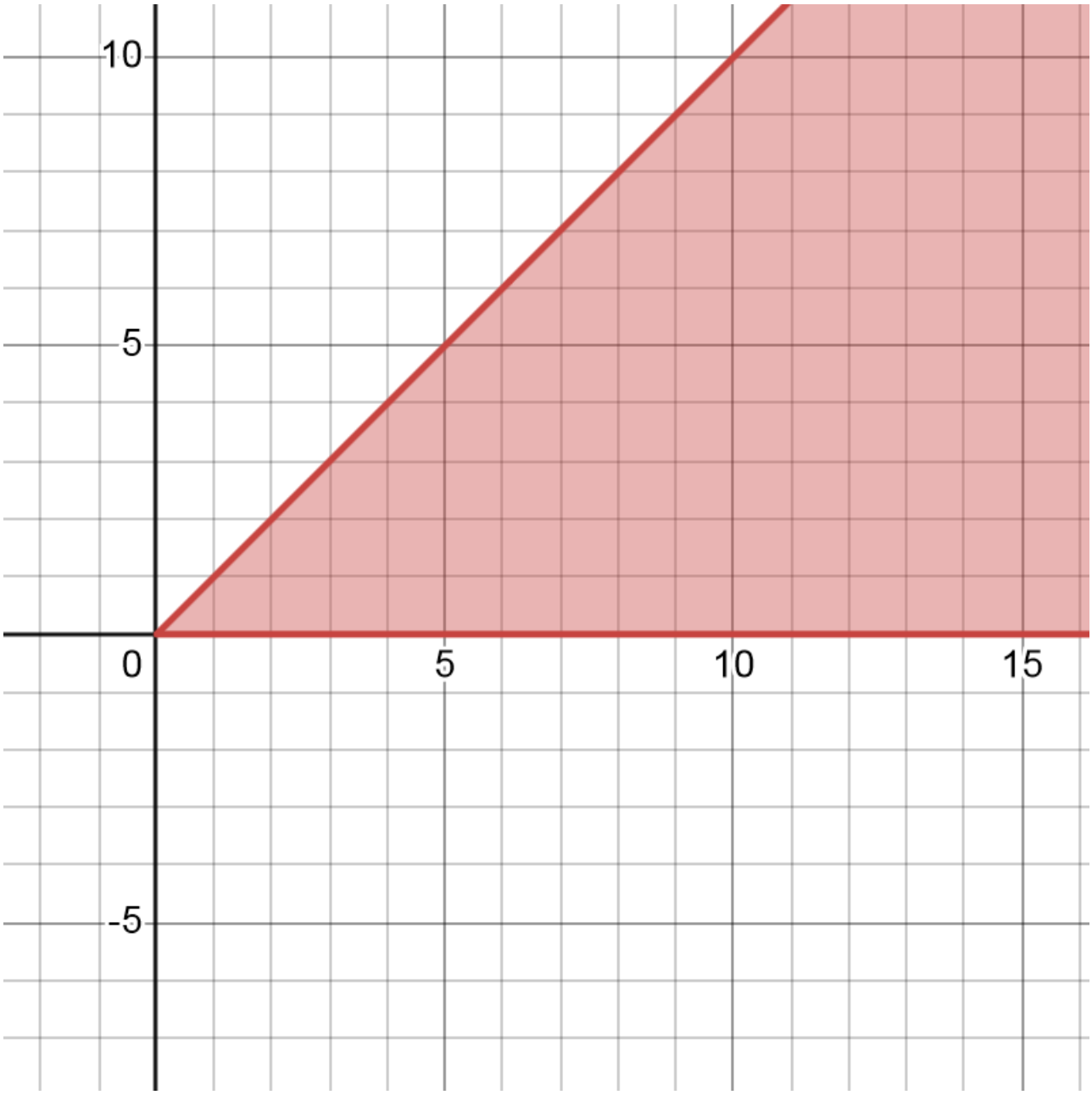
Definition

A set $C \subseteq \mathbb{R}^n$ is a **cone** if $\lambda\vec{x} \in C \ \forall\lambda \geq 0$ and $\forall\vec{x} \in C$. This is similar to the definition of convex sets, but far simpler. We only need to consider one vector $\vec{x} \in S$ instead of two vectors. We also have $\lambda \geq 0$, rather than $\lambda \in [0, 1]$.

Examples

Ex 1)

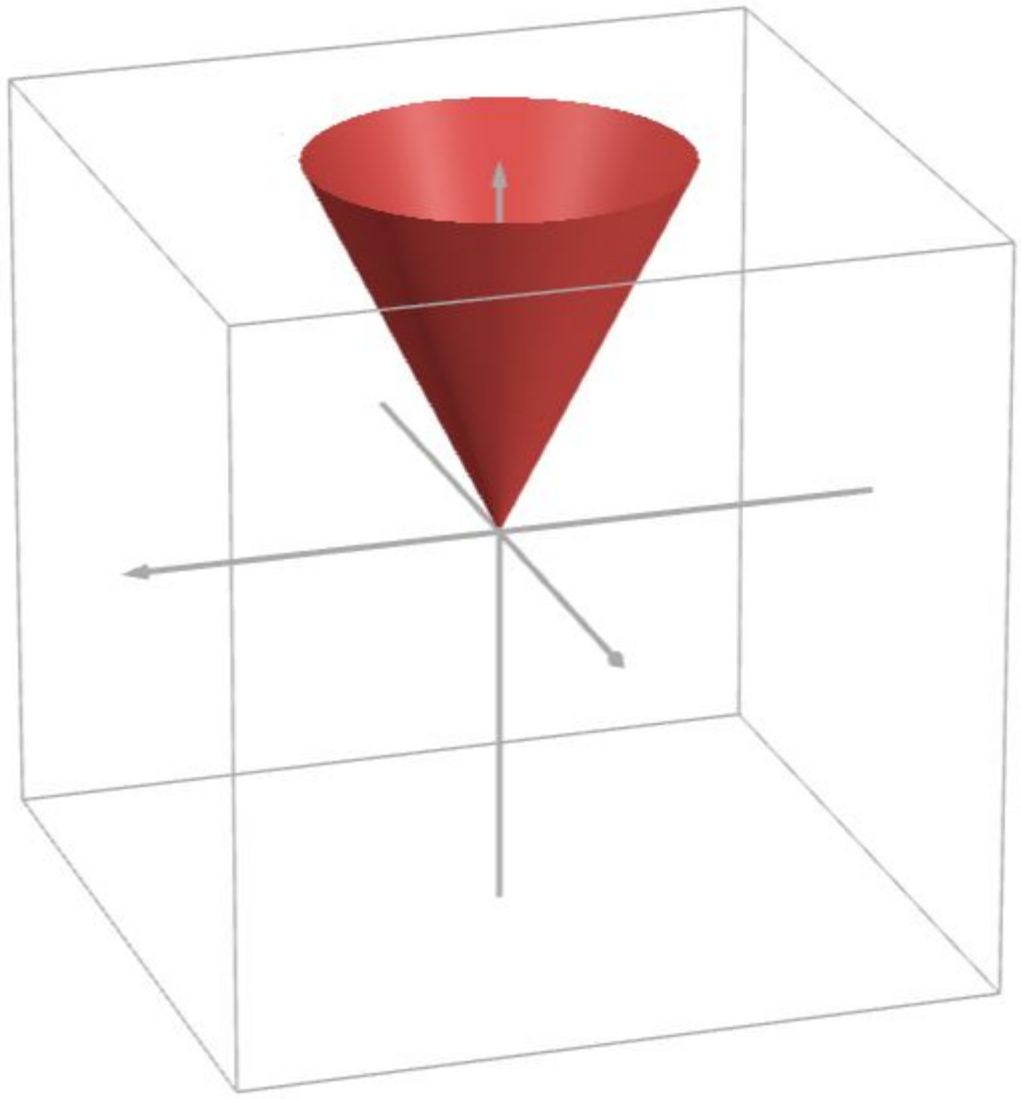
Consider the set S pictured below:



(Suppose S continues on forever along these bounds)

Ex 2)

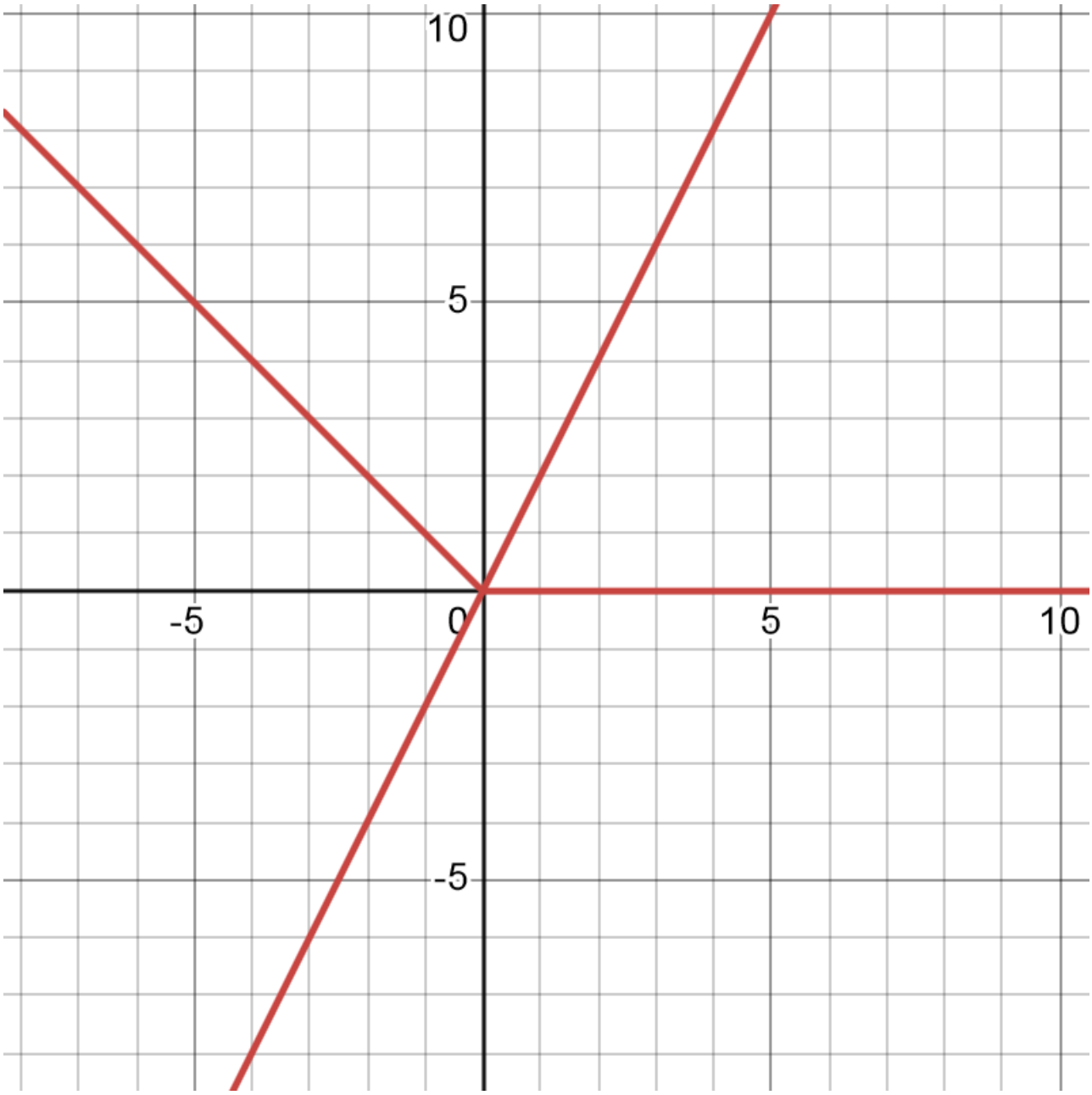
Consider the set S pictured below:



(Suppose S continues on forever along these bounds)

Ex 3)

Consider the set S pictured below:



(Suppose S continues on forever along these bounds)

Ex 4)

Consider the half space:

$$C = \{\vec{x} \in \mathbb{R}^n | \vec{a} \cdot \vec{x} \geq 0\}$$

Proof

Let $\vec{x} \in C$. Then $\vec{a} \cdot \vec{x} \geq 0$. Let $\lambda \geq 0$. Then

$$\vec{a} \cdot (\lambda \vec{x}) = \lambda \vec{a} \cdot \vec{x} \geq \lambda 0 = 0$$

Thus

$$\lambda \vec{x} \in C$$

Ex 5)

Let $C = V \leq \mathbb{R}^n$ be a subspace of $\mathbb{R}^n$. Then C is a cone.

Proof

Let $\vec{v} \in V$. Then $\lambda\vec{v} \in V$ for all $\lambda \geq 0$ as V is a vector space.

Non Ex 6)

Let P be a bounded polyhedron. P is not a cone unless $P = \{0\}$ or $P = \emptyset$.

Proof

First suppose $P = \emptyset$. Then P is a cone vacuously. Now, suppose $P = \{0\}$. Then P is a cone as

$$\lambda(0) = 0 \in P$$

for all $\lambda \geq 0$. Now, suppose P is not one of the previous two cases. Then $\exists \vec{x} \in P$ such that $\vec{x} \neq 0$. As $\vec{x} \neq 0$, then

$$0 < \|\vec{x}\|$$

We also know that P is bounded, therefore there exists $K \in \mathbb{R}$ such that

$$\|\vec{y}\| < K \ \forall \vec{y} \in P$$

Suppose $\lambda > \frac{K}{\|\vec{x}\|} > 0$. Then

$$\|\lambda\vec{x}\| = \lambda \|\vec{x}\| > \frac{K}{\|\vec{x}\|} \|\vec{x}\| = K$$

so $\|\lambda\vec{x}\| \notin P$.

Ex 7)

Let C_i be a cone for each $i \in I$ (for some indexing set I). Then

$$C = \cup_{i \in I} C_i$$

is a cone.

Proof

Suppose $\vec{x} \in C$. Then $\exists i \in I$ such that $\vec{x} \in C_i$. As C_i is a cone, then for any $\lambda \geq 0$, we have $\lambda\vec{x} \in C_i$, thus

$$\lambda\vec{x} \in \cup_{i \in I} C_i = C$$

and thus C is a cone.

Ex 8)

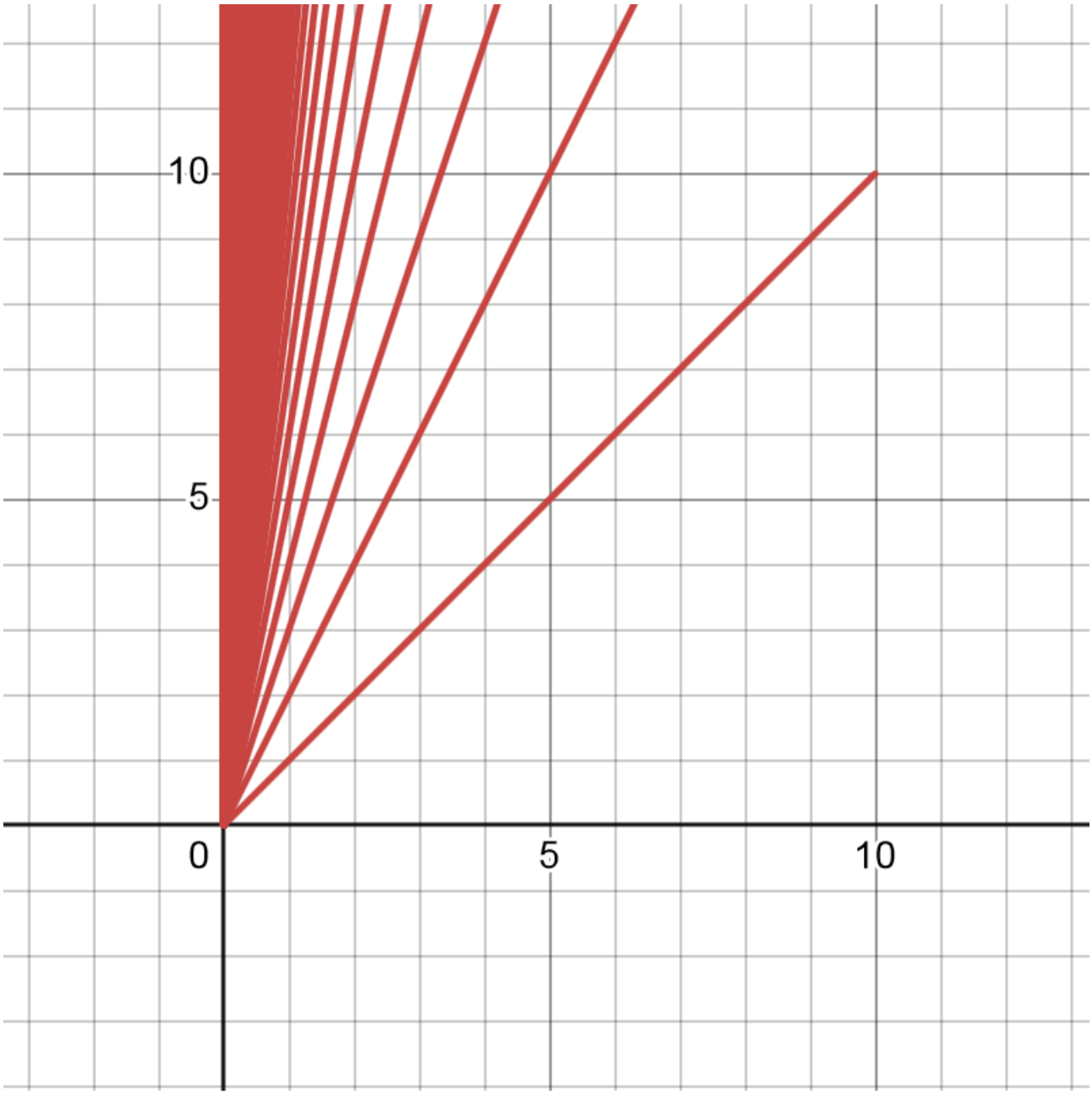
Not all cones are polyhedra! Consider the following collection of cones:

$$C_i = \left\{ \lambda\vec{x} | \lambda \geq 0, \vec{x} = \begin{pmatrix} 1 \\ i \end{pmatrix} \right\}$$

Then

$$C = \cup_{i=1}^{\infty} C_i$$

is a cone by **Ex 7)**, but C is not a polyhedron since it cannot be described by taking the intersection of finitely many inequalities.



This is a sketch of the union of the first 1000 cones. The full set cannot be drawn exactly on a computer. This example also shows that cones are not convex in general.

Conical Combinations

Let $S \subseteq \mathbb{R}^n$.Then a sum

$$\sum_{i=1}^{m} \alpha_i \vec{x}_i$$

is called a conical combination in S if each $\alpha_i \geq 0$, and each $\vec{x}_i \in S$. We specifically require that this collection of vectors $\vec{x}_i$ be finite. S need not be finite. The collection of all possible conical combinations is called the **conical hull** of S. In set notation, we denote this as

$$\operatorname{coni}(S) = \left\{ \sum_{i=1}^{k} \alpha_i \vec{x}_i | \vec{x}_i \in S, \alpha_i \geq 0, k \in \mathbb{N} \right\}$$

Theorem

The conical hull is a cone and a convex set.

Proof

Let $\lambda \geq 0$ and let $\vec{x} \in \operatorname{coni}(S)$. Then $\vec{x}$ is of the form

$$\vec{x} = \sum_{i=1}^{k} \alpha_i \vec{x}_i,\ \alpha_i \geq 0, \vec{x}_i \in S\ \forall i$$

So for $\lambda \geq 0$, we see

$$\lambda\vec{x} = \lambda \sum_{i=1}^{k} \alpha_i \vec{x}_i = \sum_{i=1}^{k} \lambda\alpha_i \vec{x}_i$$

and we see $\lambda\alpha_i \geq 0, \vec{x}_i \in S$ for all i. Thus $\lambda\vec{x} \in \operatorname{coni}(S)$. Now, to show $\operatorname{coni}(S)$ is convex. Suppose $\vec{x}, \vec{y} \in \operatorname{coni}(S)$. Then $\vec{x}, \vec{y}$ can be written in the following forms:

$$\vec{x} = \sum_{i=1}^{m} \alpha_i \vec{x}_i,\ \vec{y} = \sum_{i=1}^{k} \beta_i \vec{y}_i$$

with $\alpha_i, \beta_i \geq 0$ for all i and with $\vec{x}_i, \vec{y}_i \in S$ for all i. Let $z_i = \begin{cases} \vec{x}_i & 1 \leq i \leq m \\ \vec{y}_{i-m} & m+1 \leq i \leq m+k \end{cases}$, let $\alpha_i' = \begin{cases} \alpha_i & 1 \leq i \leq m \\ 0 & \text{otherwise} \end{cases}$ and let $\beta_i' = \begin{cases} 0 & 1 \leq i \leq m \\ \beta_{i-m} & m+1 \leq i \leq m+k \end{cases}$. Then

$$\vec{x} = \sum_{i=1}^{m+k} \alpha_i' \vec{z}_i,\ \vec{y} = \sum_{i=1}^{k+k} \beta_i' \vec{z}_i$$

So for any $\lambda \in [0, 1]$

$$\lambda\vec{x} + (1-\lambda)\vec{y} = \lambda \sum_{i=1}^{m+k} \alpha_i' \vec{z}_i + (1-\lambda) \sum_{i=1}^{m+k} \beta_i' \vec{z}_i = \sum_{i=1}^{m+k} (\lambda\alpha_i' + (1-\lambda)\beta_i')\vec{z}_i \in \operatorname{coni}(S)$$

as $\lambda\alpha_i' + (1-\lambda)\beta_i' \geq 0$ for all i and $\vec{z}_i \in S$ for all i.

Cone Properties

Here are some useful properties of cones).

1)

Let C_i be a cone for each $i \in I$ (Where I is some indexing set). Then

$$C = \cap_{i \in I} C_i$$

is also a cone.

2)

Let C_1 and C_2 be cones. Then

$$C_1 + C_2 = \{\vec{x} + \vec{y} | \vec{x} \in C_1, \vec{y} \in C_2\}$$

is also a cone.

3)

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Let C be a cone. Then

$$T(C) = \{T(\vec{x}) | \vec{x} \in C\}$$

is also a cone.

4)

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Let C be a cone. Then

$$T^{-1}(C) = \{\vec{x} | T(\vec{x}) \in C\}$$

5)
Let C_1 and C_2 be cones. Then

$$C_1 \times C_2 = \{(\vec{x}, \vec{y}) | \vec{x} \in C_1, \vec{y} \in C_2\}$$

is also a cone.

Proof

1)
Let $\vec{x} \in C$. Also, let $\lambda \geq 0$. Then

$$\vec{x} \in C_i$$

for all $i \in I$, thus

$$\lambda\vec{x} \in C_i$$

for all $i \in I$ as we know that each C_i is a cone, thus

$$\lambda\vec{x} \in \cap_{i \in I} C_i$$

2)
Let $\vec{x} + \vec{y} \in C_1 + C_2$. Let $\lambda \geq 0$. Then

$$\lambda(\vec{x} + \vec{y}) = \lambda\vec{x} + \lambda\vec{y} \in C_1 + C_2$$

as $\lambda\vec{x} \in C_1$ and $\lambda\vec{y} \in C_2$.
3)
Left as homework exercise.
4)
Let $\vec{x} \in T^{-1}(C)$ and $\lambda \geq 0$. Suppose $T(\vec{x}) = \vec{y}$. Then

$$T(\lambda\vec{x}) = \lambda T(\vec{x}) = \lambda\vec{y}$$

And as $\lambda\vec{y} \in C$, then $\lambda\vec{x} \in T^{-1}(C)$
5)
Left as homework exercise.